

DAX aggregations

Creating new tables with DAX

Creating new tables in Power BI

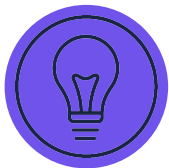
Power BI allows us to create datasets using fundamental **table creation and data modelling techniques** to suit the needs of dashboards and reports.

The careful use of DAX functions empowers us to **create tables** that **aggregate, enrich, optimise, simplify, segment, and integrate** data.

These techniques enhance our ability to design **optimised data models**, enabling us to **derive valuable business intelligence** from datasets.

Creating new tables in Power BI

There are a variety of reasons why we might want to build new tables in Power BI.
Here are a few examples:



Data aggregation and summarisation

Generate **summarised views** of data based on different **dimensions and criteria**.



Data transformation

Shape and restructure existing data to meet specific reporting or analysis requirements.



Data integration and consolidation

Consolidate data from **multiple sources** into a **single unified view** for a more comprehensive data model.



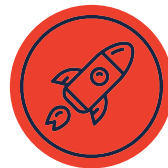
Performance optimisation

Optimise the performance of reports, especially for **complex calculations** that are **resource-intensive**.



Simplifying report design

Simplify the report design process, **reducing redundant calculations** and **improving report loading and refresh** times.



Data enrichment

Combine tables, adding **calculated columns** or **merging data** from different sources to create more comprehensive reports.

Creating new tables in Power BI

To create a new table in Power BI:

01. Navigate to the **"Table tools"** tab and select **"New table"** under the **"Calculations"** section.

02. Enter the appropriate **DAX functions** in the **formula bar** to create the new table.

03. Press **Enter** and the new table will appear in the table view.

The example datasets used in this train may be found [here](#).



The screenshot shows the Power BI Desktop interface. The 'Table tools' tab is selected in the ribbon. Under the 'Calculations' group, the 'New table' icon is highlighted with a yellow box and labeled '01.'. The formula bar at the top contains the DAX formula: `NewTable = SELECTCOLUMNS( 'Gender_parity', "Country_ID", 'Gender_parity'[Country_ID], "Indicator_Value", Gender_parity[Indicator_Value] )`. This formula bar is highlighted with a yellow box and labeled '02.'. Below the formula bar, a table view is shown with two columns: 'Country_ID' and 'Indicator_Value'. The table contains 10,469 rows, with the first few rows visible, all having 'GHA' in the 'Country_ID' column. This table view is highlighted with a yellow box and labeled '03.'. The status bar at the bottom indicates 'Table: NewTable (10,469 rows)'.

Data aggregation and summarisation

The SUMMARIZE and GROUPBY functions offer a comprehensive dataset view by **creating summary tables based** on specified dimensions. They both produce similar results.

$f(x)$ = `SUMMARIZE/GROUPBY(table_name, Groupby_Column1, Groupby_Column2, ..., "Column_name", Aggregate_function(Column1), Aggregate_function(Column2), ...)`

Summarise the **Sales** table to obtain a new table showing the **total sales for each product**.

Sales		
Product	Year	Amount
A	2021	1000
B	2021	1500
A	2022	1200



```
SummaryTable =  
SUMMARIZE('Sales', 'Sales'[Product],  
'Sales'[Year], "TotalSales",  
SUM('Sales'[Amount]))  
  
GROUPBY('Sales', 'Sales'[Product],  
'Sales'[Year], "TotalSales",  
SUM('Sales'[Amount]))
```

SummaryTable	
Product	TotalSales
A	2200
B	1500

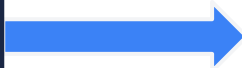
Data transformations: SELECTCOLUMNS function

SELECTCOLUMNS is used to create a new table by **selecting specific columns** from an existing table. This aids in simplifying data for clearer analysis.

$f(x)$ = `SELECTCOLUMNS(table, "New_Column_Name1", Column1, "New_Column_Name2", Column2, ...)`

Create a new table by selecting only the '**Product**' and '**Amount**' columns from the original sales data.

Sales		
Product	Year	Amount
A	2021	1000
B	2021	1500
A	2022	1200



```
SelectedColumns =  
SELECTCOLUMNS( 'Sales',  
"Product",  
'Sales'[Product],  
"Amount", 'Sales'[Amount])
```

SelectedColumns	
Product	Amount
A	1000
B	1500
A	1200

Data transformations: ADDCOLUMNS function

The ADDCOLUMNS function allows for the **addition of calculated columns** to a table. This is beneficial when new values or metrics need to be derived from existing data.

$f(x)$ = `ADDCOLUMNS(table, "New_Column_Name", Expression)`

Create a new table by adding a '**Profit**' column to the Sales table, calculated as the difference between '**Amount**' and '**Cost**'.

Sales			
Product	Year	Amount	Cost
A	2021	1000	800
B	2021	1500	1200
A	2022	1200	950

ExtendedSales =
`ADDCOLUMNS('Sales',
"Profit",
'Sales'[Amount] -
'Sales'[Cost])`

ExtendedSales				
Product	Year	Amount	Cost	Profit
A	2021	1000	800	200
B	2021	1500	1200	300
A	2022	1200	950	250

Data transformations: GENERATE function

The GENERATE function is crucial for **generating tables based on specific conditions**, enabling the creation of dynamic and customised tables.

$f(x)$ = GENERATE(Table, RowFilter)

Generate a new table by including only the rows from the original sales data where the 'Quantity' is greater than 10.

Sales			
Product	Year	Amount	Quantity
A	2021	1000	15
B	2021	1500	20
A	2022	1200	8

FilteredSales
= GENERATE('Sales',
'Sales'[Quantity] >
10)

FilteredSales			
Product	Year	Amount	Quantity
A	2021	1000	15
B	2021	1500	20

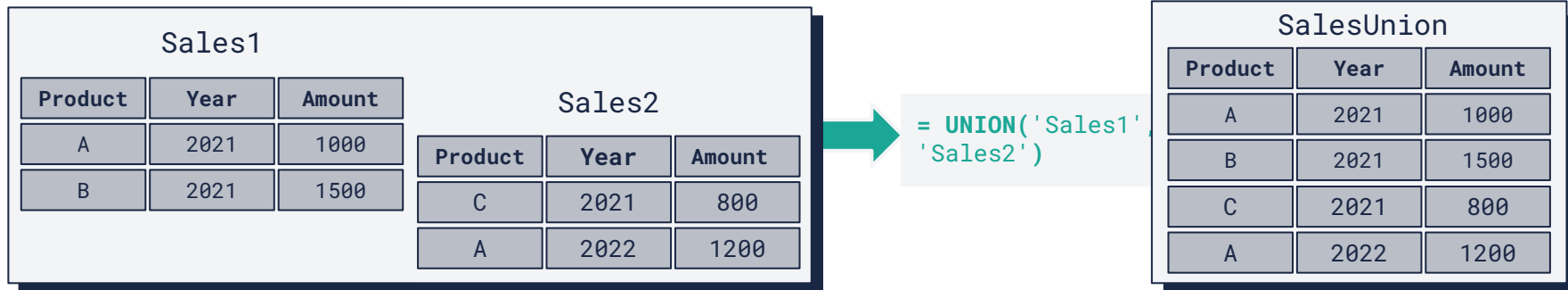
Data integration and consolidation

Functions like **UNION** and **MERGE** enable the **integration and consolidation of data from multiple tables** or sources, contributing to a more comprehensive and coherent data model.

$f(x)$ = `UNION('table1', 'table2')`

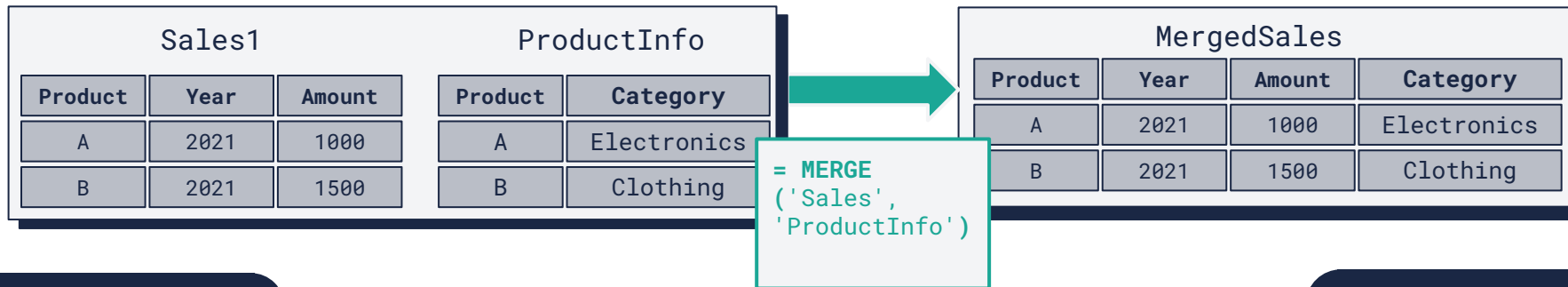
$f(x)$ = `MERGE('table1', 'table2')`

Combine the two tables, **Sales1** and **Sales2**, into a new table using the UNION function.



Data integration and consolidation

Merge the two tables ('Sales' and 'ProductInfo') into a new table based on the 'Product' column, consolidating sales data with product information.



UNION

Combines rows vertically, creating a new table with all unique rows from the combined tables.

Requires tables with the same structure (same columns and data types).

MERGE

Combines columns horizontally, creating a new table by merging columns based on a related column.

Requires tables with a common related column.

More on creating tables in Power BI

In the realm of Power BI, the creation of tables extends beyond mere data representation; it serves as a strategic lever for **optimising performance**, **simplifying report design**, and **enabling data segmentation**.

Performance optimisation

Power BI report performance can be improved by **creating calculated tables** using functions like SUMMARIZE and GROUPBY. These tables store aggregated or summarised data, **reducing the need for resource-intensive calculations** during report execution.

This **enhances report responsiveness**, especially when dealing with large datasets or complex calculations.

Simplifying report design

Creating calculated tables using DAX functions simplifies report design by **minimising redundant calculations** and **reducing the overall complexity** of the model.

This allows report designers to focus on presenting insights rather than dealing with intricate calculations, leading to **improved loading and refresh times**.

More on creating tables in Power BI

Data segmentation

Additional tables in Power BI **enable effective data segmentation** based on different dimensions, enabling users to **dynamically analyse subsets of data**, gaining deeper insights into specific business aspects.

This segmentation capability **enables more targeted and meaningful reporting**.